Now back to our question, we can reformulate it using our definitions: are all graphs planar?

The answer is, unfortunately, no: not all graphs are planar. The first algorithm to check if a graph is planar was given by the Polish mathematician Kazimierz Kuratowski; his theorem characterizes planarity in terms of *forbidden graphs*: it states, in fact, that for a graph to be planar, it can't contain two specific non-planar graphs as its subgraphs.

These two graphs are the simplest non-planar graphs are the complete graph K_5 and the complete bipartite graph $K_{3,3}$; Kuratowski's theorem states that "a graph is planar if and only if it doesn't contain as a subgraph neither K_5 nor $K_{3,3}$, nor any *subdivision* of those two graphs".

This was an amazing result, but to appreciate it better, we should first give a few more definitions.

15.1.2 Complete and Bipartite Graphs

A **complete graph** is a graph where each vertex is connected by an edge to each other vertex in the graph; in these graphs, the number of edges is maximal for simple graphs, being quadratic with respect to the number of vertices: $|E| = O(|V|^2)$.